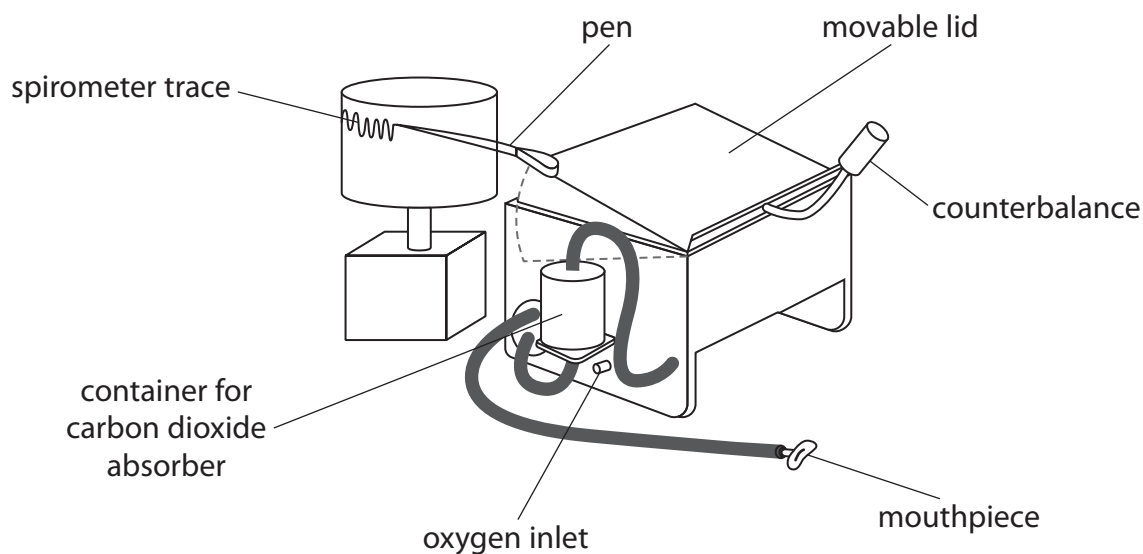


- 2 The diagram shows a spirometer that can be used to investigate human lung function.



A student used a spirometer to investigate the consumption of oxygen of a group of volunteers.

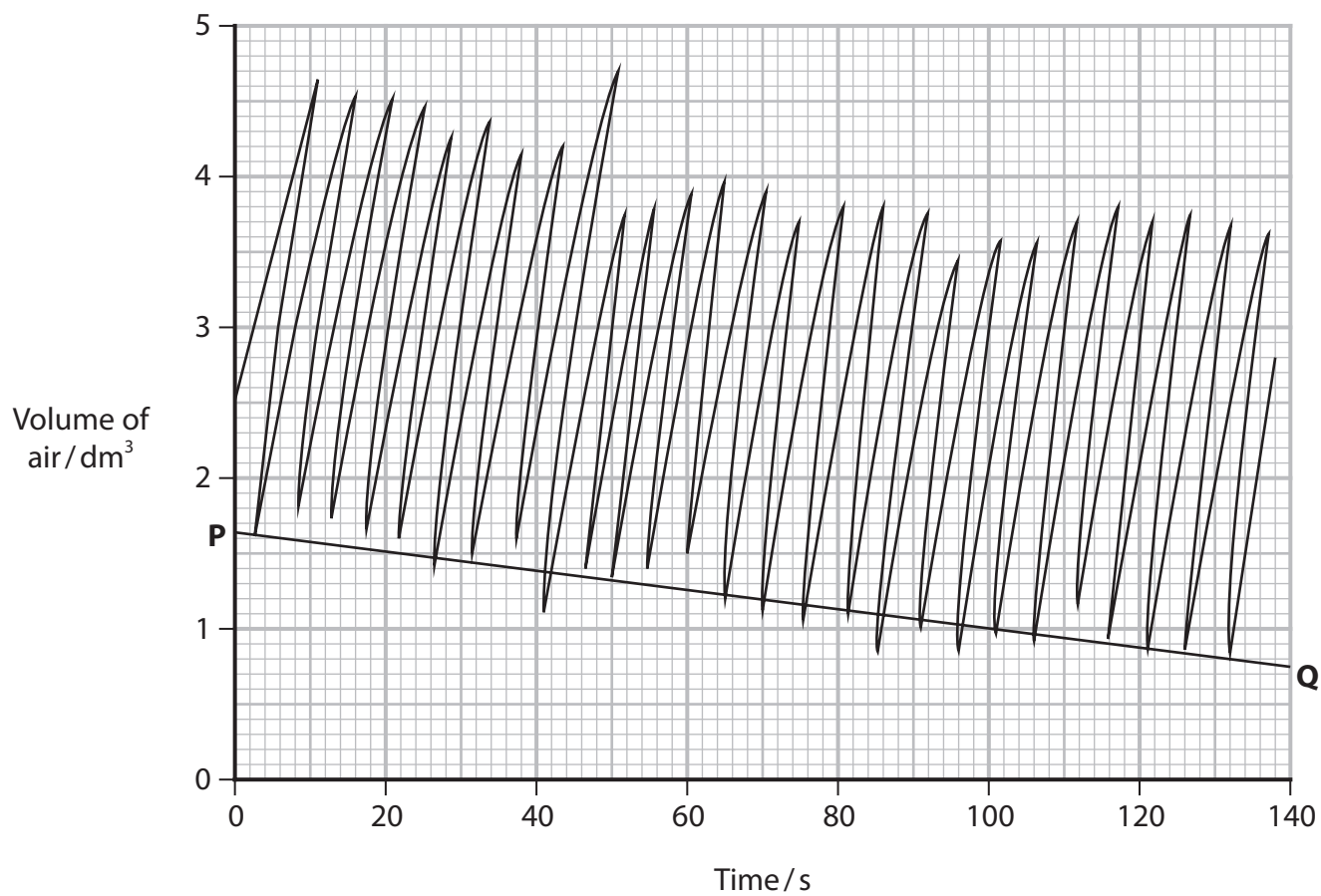
- (a) Suggest **one** risk the volunteer subjects might encounter when using this spirometer and how the student could reduce the risk.

(2)

Risk

How the student could reduce the risk.

(b) The graph shows a spirometer trace from one volunteer breathing at rest.



The line PQ indicates the oxygen consumption.

Calculate the rate of oxygen consumption between P and Q.

Give your answer with appropriate units.

(2)

Answer

(c) The student controlled several variables in this investigation.

(i) State **two biotic** variables that could affect this investigation.

(2)

First variable

Second variable

(ii) Choose **one** of the variables you have identified in (i).
Describe how this variable can be controlled.

(1)

Variable

Method of control

(d) Describe how the ventilation centre in the medulla oblongata controls the ventilation rate.

(3)

(Total for Question 2 = 10 marks)